available at www.sciencedirect.com

ScienceDirect

www.elsevier.com/locate/brainres

BRAIN RESEARCH

Research Report

Branching patterns for arterioles and venules of the human cerebral cortex

Francis Cassot[a,*], *Frederic Lauwers*[a], *Sylvie Lorthois*[b], *Prasanna Puwanarajah*[c], *Valérie Cances-Lauwers*[d], *Henri Duvernoy*[e]

[a]*Functional Neuroimaging Laboratory, INSERM U825, CHU Purpan, 31059 Toulouse-cedex 3, France*
[b]*Fluid Mechanics Institute of Toulouse, UMR CNRS/INPT/UPS 5502, 1, all. Prof. Camille Soula, 31400 Toulouse, France*
[c]*Nuffield Department of Surgery, John Radcliffe Hospital, Oxford OX3 9DU, UK*
[d]*Department of Epidemiology and Public Health, Faculty of Medicine, Toulouse, France*
[e]*12 chemin des relançons, 25 000 Besançon, France*

ARTICLE INFO

Article history:
Accepted 2 December 2009
Available online 11 December 2009

Keywords:
Microcirculation
Human brain
Vascular tree
Bifurcation

ABSTRACT

Branching patterns of microvascular networks influence vascular resistance and allow control of peripheral flow distribution. The aim of this paper was to analyze these branching patterns in human cerebral cortex. Digital three-dimensional images of the microvascular network were obtained from thick sections of India ink-injected human brain by confocal laser microscopy covering a large zone of secondary cortex. A novel segmentation method was used to extract the skeletons of 228 vascular trees (152 arterioles and 76 venules) and measure the diameter at every vertex. The branching patterns (area ratios and angles of bifurcations) of nearly 10,000 bifurcations of cortical vascular trees were analyzed, establishing their statistical properties and structural variations as a function of the vessel nature (arterioles versus venules), the parent vessel topological order or the bifurcation type. We also describe their connectivity and discuss the relevance of the assumed optimal design of vascular branching to account for the complex nature of microvascular architecture. The functional implications of some of these structural variations are considered. The branching patterns established from a large database of a human organ contributes to a better understanding of the bifurcation design and provides an essential reference both for diagnosis and for a future large reconstruction of cerebral microvascular network.

© 2009 Elsevier B.V. All rights reserved.

* *Corresponding author.*
E-mail address: francis.cassot@inserm.fr (F. Cassot).

Abbreviations: BDI, Branching Development Index; ALB, asymmetric-lateral bifurcation; HSB, homogeneous-symmetrical bifurcation; IEB, inter-element bifurcation; OAI, Order Asymmetry Index (i.e., the difference between the parent vessel order and the smaller daughter branch order); A2/A1, asymmetry ratio; AR, area ratio; A1/A0, major area ratio, between cross-sectional areas of the mother and the larger daughter branch; A2/A0, minor area ratio, between the cross-sectional areas of the mother and the smaller daughter branch; AER, area expansion ratio or total area ratio (A1+A2)/A0; γ=A2/A1, asymmetry ratio; $\alpha 01$, angle between the mother and the larger daughter branch (feed continuation angle); $\alpha 02$, angle between the mother and the smaller daughter branch; $\alpha 12$, angle between both daughter branches; β, asymmetry angle, $\alpha 01-\alpha 02$; $\pi-\Omega$, out-of-plane angle, planarity of the bifurcation, $\Omega=\alpha 01+\alpha 02+i12-\pi$; SD, standard deviation; CSD, circular standard deviation; IQR, interquartile range

0006-8993/$ – see front matter © 2009 Elsevier B.V. All rights reserved.
doi:10.1016/j.brainres.2009.12.007

1. Introduction

Vascular architecture, particularly of cerebral microvessels, has profound implications for both health and disease. Indeed, providing accurate and comprehensive measurements of vascular morphology is an essential step in the analysis of vascular formation as well as of the structural adaptation of microcirculation under physiological (development, growth and exercise) or pathophysiological conditions (inflammation, hypertension, diabetes, obesity and metabolic syndrome) (Jonk et al., 2007; Levy et al., 2008; Murfee and Schmid-Schonbein, 2008; Patton et al., 2005; Pries et al., 2005; Wiernsperger et al., 2007). More generally, "almost every disease from cancer to the common cold affects blood vessel attributes (vessel number, radius, tortuosity, and branching pattern)" (Bullitt et al., 2005a).Vascular alterations associated with pathology have gained increased attention (Lu et al., 2004) and pathological angiogenesis is now recognized as a hallmark of disease (Carmeliet and Jain, 2000; Carmeliet, 2005; Fukumura and Jain, 2008). Furthermore, the cerebral microcirculation is linked to a number of areas in neuropathology including neurological disease such as Alzheimer's (Meyer et al., 2008), vascular occlusions in cerebro-vascular disease (Nishimura et al., 2007), blood vessel morphologic changes during development and treatment of brain tumors (Baish et al., 1996; Baish and Jain, 2000; Bullitt et al., 2005b, 2007; Jain, 2005).

The analysis of these states using functional neuroimaging clearly also requires a sound, underpinning knowledge of both the macro- and microcirculations. In functional magnetic resonance imaging (fMRI), the usefulness of the blood oxygen level dependent (BOLD) technique as a surrogate signal strongly depends on an understanding of neurovascular coupling and the underlying vascular architecture (Goense et al., 2007; Vanzetta, 2006). One established group in this field concluded recently that "tomographic assessments of the vasculature in large cortical fields of view are necessary to obtain the three-dimensional topology of the vascular network, using the currently available methods" (Weber et al., 2008).

Thus, the morphological, topological and functional study of cerebral microcirculation is a topic of growing interest to both vascular physiologists and neuroradiologists (Cassot et al., 2006; Heinzer et al., 2008; Lauwers et al., 2008; Risser et al., 2007; Weber et al., 2008).

Our group recently developed a 3D computer-assisted method for microvascular cerebral network analysis and gathered a quantitative data library on the architecture of the microcirculation of the human cerebral cortex (Cassot et al., 2006). In particular, this large database was analyzed statistically to assess the general morphometric and topological characteristics of the arteriolar, venular and capillary cortical networks, as well as the variations of microvascular density as a function of cortical depth (Lauwers et al., 2008). Moreover, an analysis of the branching vessels of these networks showed that there were scaling laws governing the relationships between cumulative length, cumulative volume and the flow through these vessels(Cassot et al., 2009), giving further evidence of their fractal nature and experimental support for allometric equations (West and Brown, 2005).

In this paper, we subject the data to further analysis with emphasis on properties of bifurcations, specifically area ratios, branching angles and junction exponents. Indeed, anatomists or clinicians focused on the local geometrical properties either of bifurcations, such as optimality deviance (Tapp et al., 2007; Witt et al., 2006), or of the vascular segments themselves, such as tortuosity, and showed their clinical interest. However, most of these previous studies have been rather qualitative, or even subjective, and based on a relatively small number of measurements of the vessel attributes under consideration. Moreover, many of these analyses were derived from 2D measurements, the inadequacy of which has been demonstrated by subsequent 3D data (O'Flynn et al., 2007). Ultimately, little is known about the statistical properties of vessel attributes or branching patterns. In particular, it is unclear whether vessel attributes fit a Gaussian distribution and how dependent upon anatomical location these values might be (Bullitt et al., 2005a). In spite of recent attempts to define better indices to represent the attribute values of a vessel population, which could be quantified automatically



Fig. 1 – Arterioles (grey) and venules (black). Scale bar = 1 mm.

and lead to robust characterization, and advances in our understanding of vascular architecture (Heinzer et al., 2008), work is still necessary to reach these goals providing answers to key clinical questions. Revealing "signatures" of normal vascular trees would provide a rationale for diagnosis purpose and would therefore be of value.

Besides this clinical interest, the study of these local properties has also important functional implications as demonstrated in a recent paper of Kassab's group (Kaimovitz et al., 2008), for coronary network, and that we will discuss further in this paper, for brain microvascular network. Actually, phase separation effect, a local phenomena governed by the branching pattern, induces a first coupling between architecture and hemodynamics. Moreover, structural adaptation of the microcirculation as well as the vessels growth process are governed, among other things, by the spatial distribution of pressure throughout the whole microvascular network, which is inherently nonlocal; i.e., the pressure at a given point in space is influenced not only by the pressure distribution in its immediate neighborhood but more globally by the pressure distribution in the entire network. Thus, all the details of the vascular geometry, including the capillary net, the arterial, and venous trees, need to be considered in order to analyze the flow field, particularly the wall shear-stress distribution, in the organ. Finally, this local characterization is important for the future construction of a realistic hemodynamic network model.

Thus, in this paper, we focus on the branching patterns (area ratios and angles of bifurcations) of cortical vascular trees, establishing their statistical properties as a function of the vessel nature (arterioles versus venules), of their taxonomy (based on the parent vessel hierarchical position) or of their type. We also describe their connectivity and discuss whether the assumed optimal design of vascular branching can account for the complex reality of microvascular architecture.

2. Results

2.1. General data on extracted vascular trees

In this first section of the results, we present the global statistics for the morphometric data under consideration. Two hundred twenty-eight vascular arborescences were extracted from 10 mosaics covering a surface of approximately 1.5 cm^2 and encompassing a cerebral tissue volume of 28.6 mm^3. They were divided into 152 arterial and 76 venous trees according to their morphological features. Fig. 1 displays some examples of these arborescences. These vessels were segmented in 7604 arterial elements (11014 arterial segments[1]) and 5592 venous elements (8042 venous segments). In this population, we observed 9414 nodes (bifurcations): 5431 arterial and 3983 venous. The mean number of nodes by arborescence was then approximately 41, varying from less than 36 for arterioles to more than 52 for venules displaying therefore a significant increase of almost 50% of node density for veins compared to arterial trees.

Table 1 – Relative population sizes (%) depending on the nature (arterioles, venules), the parent vessel order or the type of the bifurcations (top) and as a function of the parent vessel order for the three types of bifurcations (bottom).

Group	Nature		Parent vessel order				Type		
	A	V	1	2	3	4	ALB	HSB	IEB
Size (%)	57.7	42.3	71.3	21.7	6.66	0.36	54	37.4	8.6
Type/order	1	2	3	4					
ALB	62.6	27.1	9.66	0.55					
HSB	79.5	16.6	3.7	0.17					
IEB	89.7	9.8	0.5	0					

The total number of orders between a postcapillary venule or a precapillary arteriole and its pial origin in the sulcus, referred as the branching development index (BDI), ranged from 3 to 5.

More than half of the bifurcations (54%) belongs to the asymmetric-lateral (ALB) type, in which a lower order vessel branches laterally between two segments of the same element and more than one third (37.4%) belongs to the homogenous-symmetrical one (HSB) in which the order of the two daughter branches equals that of the parent vessel minus one. The remaining (8.6%) bifurcations belong to the inter-element bifurcation (IEB) group in which the three branches have the same order. Table 1 summarizes the relative sizes of the different groups or subgroups defined on the basis of their nature, taxonomy or type.

The quantity of data collected here is the largest of its kind thus far, ensuring the significance of the study. Naturally, other cortical zones may display different anatomical patterns.

2.2. Area branching ratios

This section focuses on the four area branching ratios. We consider global features as well as statistically comparing the populations as differentiated by vascular nature, topology or bifurcation type.

2.2.1. Global statistics

We analyzed the cross-section area ratios for each bifurcation of the 228 vascular trees. Table 2 summarizes the general characteristics of these ratios. The distributions of area ratios

Table 2 – Main statistics of area ratios.

Area ratio	$(A_1+A_2)/A_0$	A_1/A_0	A_2/A_0	A_2/A_1
Median	1.519	0.905	0.608	0.721
Mean	1.626	0.984	0.642	0.686
±SD	±0.684	±0.46	±0.294	±0.213

[1] A segment is a vessel section between two successive bifurcations. An element can be made of solitary or various vascular segments with the same order (see Section 4.2) connected in series.

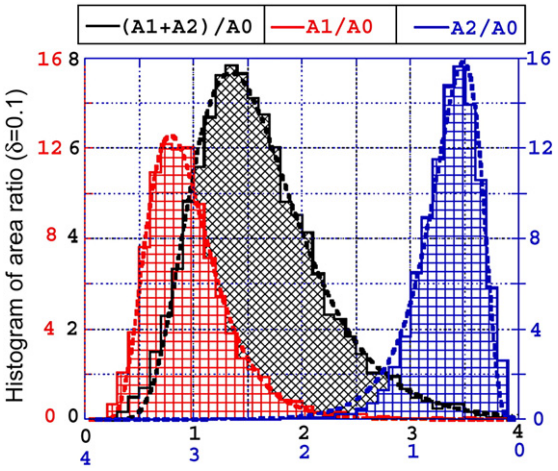


Fig. 2 – Histograms of area ratios with their respective lognormal curve fits.

are lognormal with medians equal to 1.52, 0.91 and 0.61 respectively for the total, major and minor area ratios. Fig. 2 displays the histograms of area ratios with their lognormal curve fits. The asymmetry ratio, i.e., the ratio A_2/A_1, is between 0 and 1, normally distributed with a maximal density of probability at 0.90 and a variance $\sigma^2=0.32$.

Nevertheless, it is interesting to note that nearly 40% of the bifurcations have an "inverted" profile with a cross-sectional area of the major daughter branch greater than that of the parent vessel; this inversion is also present in the minor branch but in less than 10% of the bifurcations, and for 87% of them the total cross-sectional area of the daughter vessels is greater than that of the parent branch, highlighting a shortcoming of the area conservation hypothesis.

2.2.2. Pattern differentiation with nature, taxonomy and type

Significant differences between the bifurcations area ratios are observed depending on their nature (arteriolar or venular),

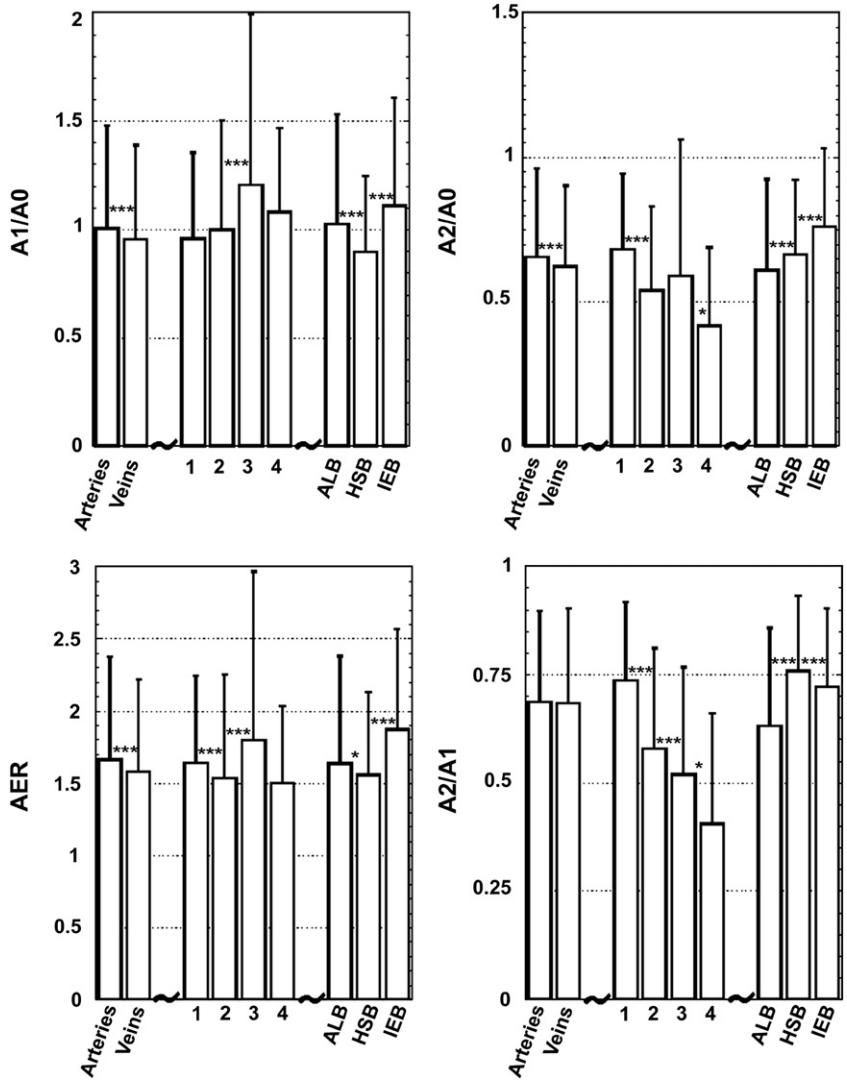


Fig. 3 – Mean values of area ratios and asymmetry ratio for arterioles and venules, as a function of the parent vessel order and for the three types of bifurcations. Error bars = standard deviation. Asterisks between consecutive mean values correspond to significance levels for the differences between the corresponding groups as assessed by non-parametric Wilcoxon–Mann–Whitney rank tests (*P < 0.05; **P < 0.001; ***P < 0.0001) (see Supplementary Table 4 for more details).

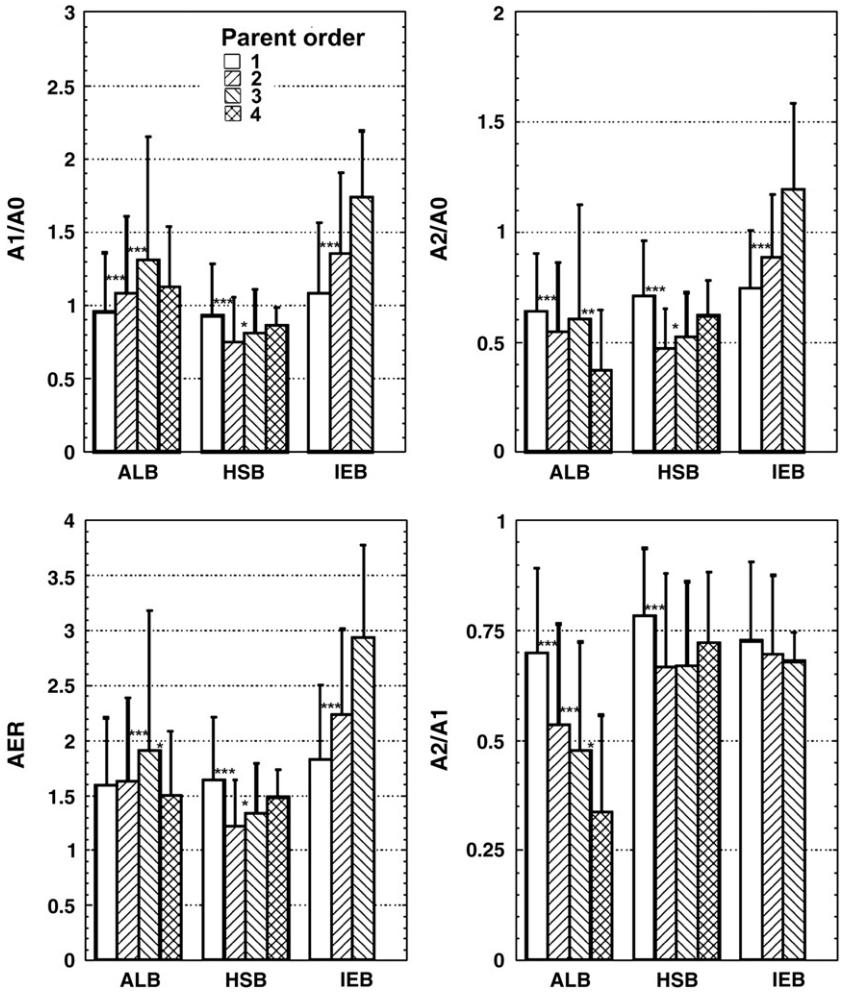


Fig. 4 – Mean values of area ratios for the three types of bifurcations as a function of the parent vessel order. Error bars = standard deviation. Asterisks between consecutive mean values illustrate significant differences as assessed by non-parametric Wilcoxon–Mann–Whitney rank tests (*P < 0.05; **P < 0.001; ***P < 0.0001) (see Supplementary Table 5 for more details).

on the taxonomic order of the parent vessel (i.e., their topological location in the branching network) or on the bifurcation type. They are illustrated on Fig. 3. The mean values of each area ratio (AR) for arterioles and venules show small but ubiquitous and statistically significant differences, the arteriolar ARs being approximately 5% higher than the venular ones, except for the asymmetry ratio which takes nearly identical values for both groups. The relationships between the ARs and the parent vessel order show a more varied pattern: a monotonic trend is observed for the major and minor ARs but in opposite directions specifically an increase for the branching ratio of the larger daughter branch and a decrease for that of the smaller daughter branch. As a consequence of these opposing trends, the relationship between the AER and the parent vessel order is no more monotonic, the AER being minimum at order 2; conversely, the asymmetry ratio decreases incrementally as the parent vessel order increases, with marked and statistically significant differences between each order group.

The mean values of ARs and asymmetry ratios for the three bifurcation types show very significant differences. The relationships between these mean values and the Order Asymmetry Index (OAI) show an incremental decreasing trend for the minor branching ratio only. The major AR and the AER show a minimum mean value at type 2 (HSB). The maximal value of the asymmetry ratio, observed for this type, infers that the asymmetry between the two daughter branches is minimal too, since the higher the asymmetry ratio, the lower the asymmetry.

However, within each type group, the relationships between the ARs and the parent vessel order are qualitatively and quantitatively different as illustrated by Fig. 4. For the first group (ALB), the opposite incremental trends of increasing major AR and decreasing minor AR result both in a consistent and markedly decreasing asymmetry ratio and a non-monotonic AER with a maximal value at order 2. In the second group

Table 3 – Mean branching angles, angular asymmetry and planarity (CSD: circular standard deviation).

Angle	A01	A02	A12	Asymmetryasym.	Solid (Ω)
Mean ± CSD	127.1 ± 27.3	114.5 ± 26	103.8 27.4	12.7 ± 42.7	166.8 ± 23.9

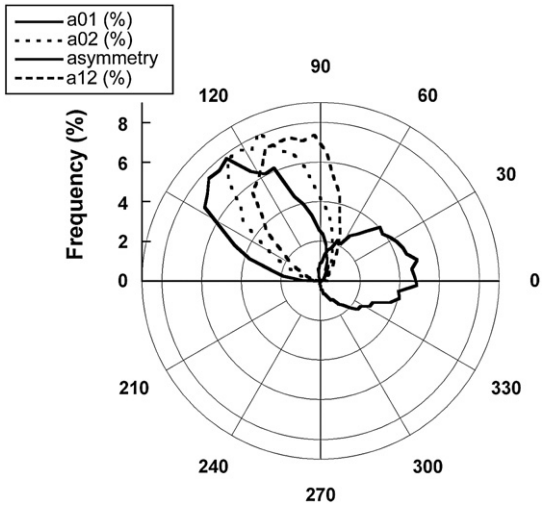


Fig. 5 – Polar histograms of bifurcation angles and angular asymmetry.

(HSB), both the ARs and asymmetry ratio behaves similarly with respect to the parent vessel order: they all take a significant maximum at the lowest order followed by a step decrease and then a progressive and weakly significant increase at higher orders. In the third group (IEB), the three ARs increase incrementally, while the asymmetry ratio tends to decrease progressively but not significantly.

In summary, we can say that differences in branching area ratios exist between arterioles and venules and that quantitative differences between branching progression within each type can also be detected according to the taxonomy employed.

2.3. *Branching angles*

A similar analysis was applied to the four branching angles and to the planarity.

2.3.1. *Global statistics*

Table 3 gives the mean values of the angles between the branches of a bifurcation; this table gives values relating both to arterioles and venules and cumulative values for all nodes. Statistical analysis of all groups under consideration reveals that the mean values of the bifurcation angle are significantly different: the angle between the mother and the larger daughter branch is greater than the angle between the mother and smaller daughter branch which itself is greater than the angle between both daughter branches ($\alpha_{01} > \alpha_{02} > \alpha_{12}$). The mean angular asymmetry between the daughter branches and the mean planarity (out-of-angle) have the same order of magnitude (approximately 13°). The angular dispersion, measured as the circular standard deviation (see Experimental Procedure) has the same order of magnitude (between 24° and 28°) for any angle of the bifurcation, except for the angular asymmetry which displays angular dispersions which are nearly twice (between 40° and 44°) that of the other branching angles. This is not surprising as this angle corresponds to the difference of the angles between the mother branch and each daughter branch.

The statistical distributions of the three main branching angles and the asymmetry (illustrated by the polar histograms of Fig. 5) are close to normal (median≈mean small skewness and excess) as suggested by the histogram of A01 and its Gaussian fit displayed on Fig. 6 (top).

Conversely, the distribution of the planarity has a large skewness and decreases steeply from a maximum corresponding to a flat (planar) bifurcation, with a quasi-hyperbolic histogram (Fig. 6, bottom). It is worth noting that nearly one third of the bifurcations can be considered flat (planarity < 2°), more than half of the bifurcations are quasi-flat (planarity < 6°), the three-quarters have a planarity less than 20° and less than 10 % have a marked planarity greater than 50°.

2.3.2. *Pattern differentiation with nature, taxonomy and type*

As for the area ratios, significant differences between the bifurcations angles are observed depending on their nature, taxonomy or type, as illustrated on Fig. 7. However, these variations are more regular than those observed for the area ratios, and when differences between groups are statistically significant, they display incremental trends.

All these angles, except the planarity, have significantly different mean values for arterioles and venules.

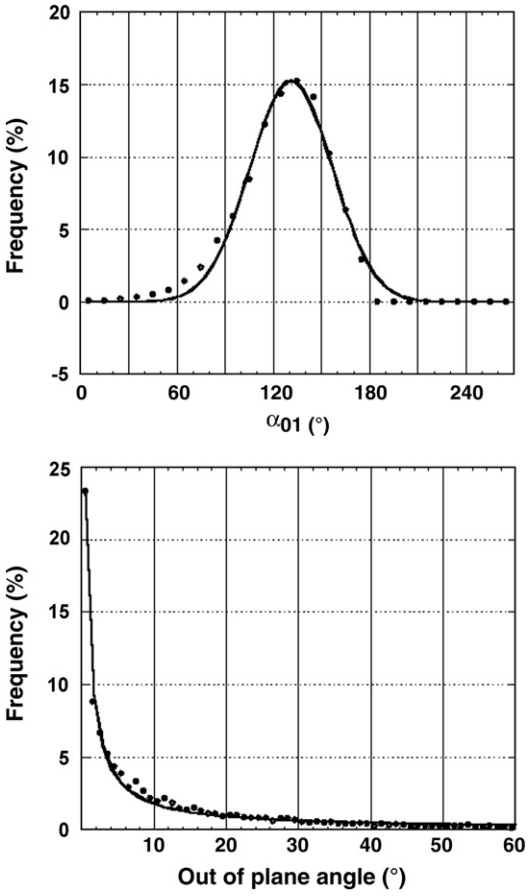


Fig. 6 – Histograms of the angle between the parent vessel and the major daughter branch (top) and of the out-of-plane angle (bottom) with the corresponding Gaussian and hyperbolic-shaped curve fits.

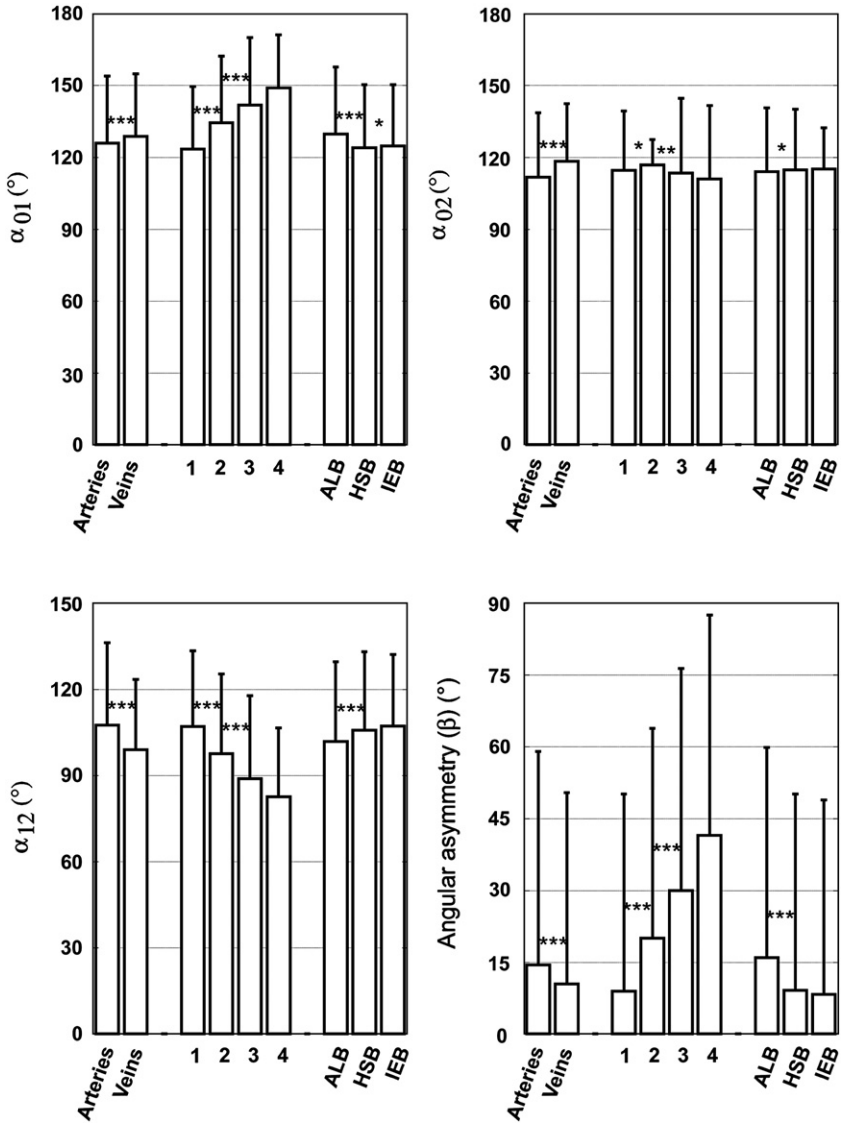


Fig. 7 – Mean values of branching angles and angular asymmetry depending on the nature, parent vessel order or typology of the bifurcations. Error bars = circular standard deviation. Asterisks between consecutive mean values correspond to significance levels for the differences between the corresponding groups as assessed by Watson's U^2 tests with ties (*P < 0.05; **P < 0.001; ***P < 0.0001) (see Supplementary Table 6 for more details).

The mean angle made by the parent vessel of a venular bifurcation with any daughter branch is greater than the corresponding angle for an arteriolar bifurcation. Of course, as a consequence, the mean angle between the daughter branches of a venular bifurcation is clearly smaller than the mean angle between the daughter branches of an arteriolar node. In addition, the mean angular asymmetry of a venular bifurcation is marginally but significantly smaller.

The relationships between the bifurcation angles and the parent vessel order may be summarized as follow: the feed continuation angle (α_{01}) increases significantly with the order; there are small or no significant differences between the mean values of the minor daughter angle (α_{02}) at any order; consequently, the angle between the daughter branches (α_{12}) decreases and the angular asymmetry significantly increases with the order.

The minor daughter angle (α_{02}) tends not to vary whatever the bifurcation type. Also, there are essentially no significant differences too between any mean angles, for the HSB and IEB types. As such, the only significant differences observed as a function of bifurcation type concern the feed continuation angle (α_{01}) of the ALB bifurcation type, which is higher than the corresponding angle of both other types which, in turn, generates a higher angular asymmetry and a lower angle between daughter branches for this type with respect to both others.

The relationships between the bifurcation angles and the parent vessel order are qualitatively very similar for ALB and HSB types as illustrated by Fig. 8. Their feed continuation (α_{01}) and asymmetry angles tend to increase significantly with the parent vessel order, while the differences between the minor daughter angles (α_{02}) are much less or not significant at all.

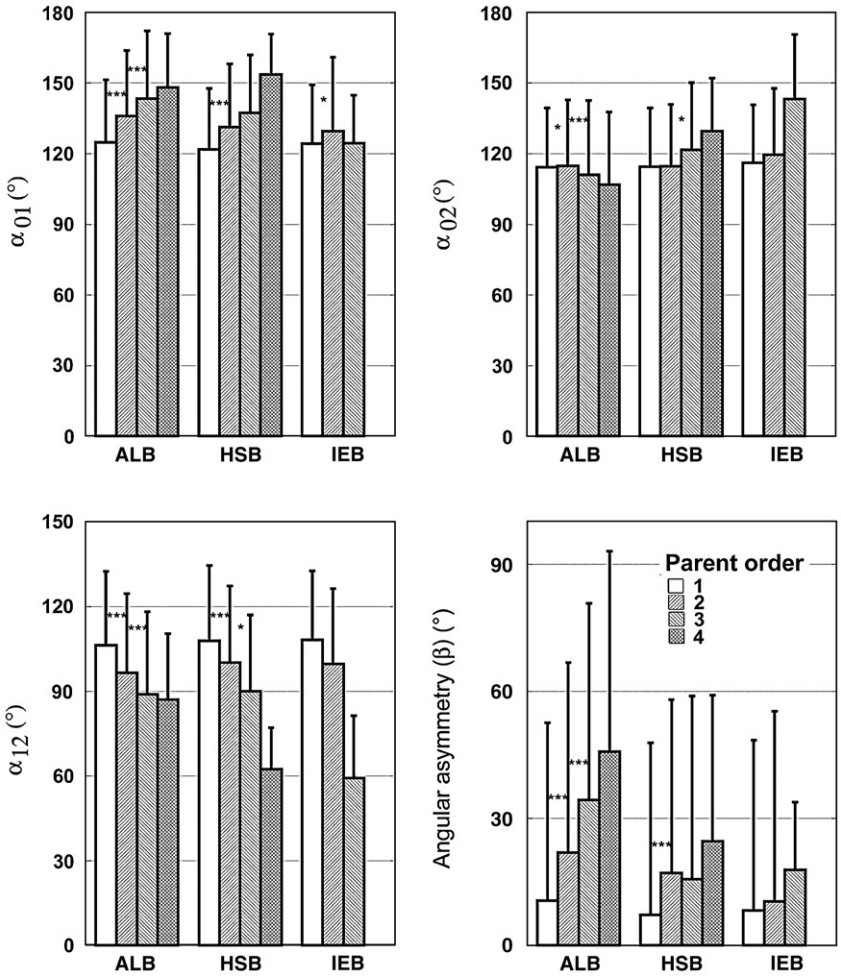


Fig. 8 – Mean values of branching angles for the three types of bifurcations as a function of the parent vessel order. Error bars = standard deviation. Asterisks between consecutive mean values illustrate significant differences as assessed by Watson's U^2 tests with ties (*$P<0.05$; **$P<0.001$; ***$P<0.0001$) (see Supplementary Table 7 for more details).

Consequently, the angles between the daughter branches (α_{12}) decrease with the parent vessel order. Within the IEB type, there are no significant differences save the mean feed continuation angle (α_{01}), which shows a weakly significant maximal value at order 2.

Finally, there are differences between the branching angles of the bifurcations populations, differentiated by their nature, taxonomy or type, as well as between the area ratios, with regular incremental variations than those observed for the branching ratios.

2.4. *Optimality of branching patterns*

In this section, we address the controversial question of the optimal design of vascular bifurcations using our database on cerebral microcirculation. We analyzed differences between the populations considered.

In a previous paper (Cassot et al., 2009), we showed the very large spread of the exponent, k, of the generalized Murray's law, as displayed on Fig. 9 (top) (median: 3.58; IQR: 2.28–6.14; 5%–95% interval: 1.36–16.33). Here we have analyzed how the median values of this parameter vary as a function of the nature, taxonomy of the parent vessel and bifurcation type (see Fig. 9, bottom): the exponent for arteriolar or venular bifurcations are not significantly different; the median exponent for a parent vessel of order 1 is close to 4 and differs significantly from the exponents of higher orders that are less than 3; and the median values of the exponent vary significantly with the type of the bifurcation and increase as the order asymmetry index decreases.

Our results, based on a far larger database than those previously used for such analyses, clearly do not support the optimality hypothesis. Moreover, if the bifurcation exponent values are similar for arterioles and venules, they should be discussed as a function of taxonomy and type.

2.5. *Connectivity*

In this last section, we consider the connectivity, according to the taxonomy defined, a key feature to consider for the construction of a network model.

Fig. 10 is a graphical representation of the connectivity matrix as a stack column plot. The column labelled j represents the column j of the connectivity matrix. In each

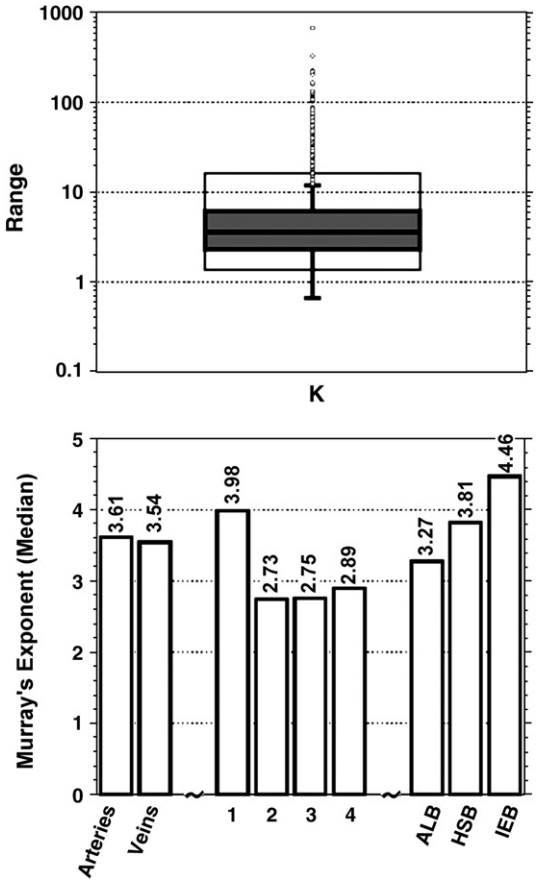


Fig. 9 – Box plot and percentiles of the bifurcation exponent *k* (top) and median values of *k* for arterioles and venules as a function of the parent vessel order and for the three types of bifurcations (bottom).

column the bottom stack corresponds to the diagonal term (*j*,*j*) of the matrix and the top stack to the (0,*j*) term of the matrix (if non-zero), i.e., to the number of daughter vessels of order 0 issued from a parent vessel of order *j* relative to the total number of parent vessels of order *j*. The intermediate stacks represent the (*i*,*j*) terms of the matrix for $i<j$. Note that the connectivity ratio of a daughter vessel issued from a parent vessel of the same order (the diagonal terms of the connectivity matrix, or the bottom stack in its graphical representation) increases very slightly, from 0.7 to 0.8, as the order of the parent vessel increases and, of course, the number of daughter vessels issued from a parent vessel of higher order (the sum of the non-diagonal terms), decreases from 1.3 to 1.2. Thus, in general terms, we can say that 4 parent vessels of a given order give rise to at least 3 daughter vessels of the same order and to 5 daughter vessels of a lower order.

3. Discussion

Areas where knowledge of microvascular architecture plays an important role are numerous and varied, with burgeoning interest in morphometric and topological parameters such as microvascular density, fractal dimension and extravascular distance. In a previous paper (Lauwers et al., 2008), we have outlined the implications for neuroimaging techniques of the varying microvascular density along the cortical layers. Among the many fields where our methods and the results obtained from them might yield useful insights are angiogenesis and development, Alzheimer's disease, vascular architecture and in evaluating the effectiveness of cancer therapeutics. Furthermore, analysis of the microvascular architecture may reveal "signatures" of normal vascular trees, allowing the subsequent diagnosis of disease processes in abnormal ones.

3.1. *Main findings of the study*

In this study, we have created a large database of the microcirculation of the human cerebral cortex from thick sections of India ink-injected human brain, using confocal laser microscopy, over a large zone of secondary cortex, and a novel method of segmentation. From this database, we separated the tree-like part from the net-like or capillary part of this vascular network. In previous papers (Cassot et al., 2006; Lauwers et al., 2008), we have highlighted the fractal nature of these branching arteriolar and venular trees and demonstrated that their topological organization is governed by scaling laws. The present study was aimed at describing the finer detail of their branching patterns. Branching area ratios, branching angles and connectivity matrix were computed at nearly 10,000 bifurcations and their statistical distributions analyzed. The main results of this study can be summarized as follows.

The area branching ratios show a lognormal distribution, but the asymmetry ratio is normally distributed within its limited range between 0 and 1. The bifurcation angles, including the angular asymmetry between daughter branches, are also normally distributed. However, the planarity is an exception to this rule, displaying a left-skewed hyperbolic statistical distribution showing that, for one third of the bifurcations, the three branches belong to the same plane, at

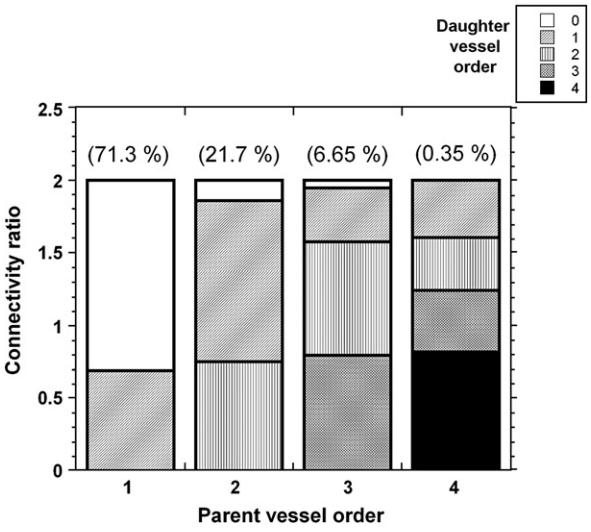


Fig. 10 – Stack column plot of the connectivity matrix. The percentage on top of each column of order *j* indicates the percentage of segments of order *j* – 1.

least in the neighborhood of the nodes, and that, in most of the cases, their directions are close to this planar profile. On average, the feed continuation angle is higher than the lateral branching angle (the angle between the parent vessel and the minor daughter branch), which is itself higher than the angle between the daughter branches, while the mean asymmetry and out-of-plane angles are far smaller.

Branching patterns show small but significant differences depending on the nature (arteriolar or venular), the taxonomy or the type of the bifurcations. In particular, the arteriolar area ratios are approximately 5% higher than that of the venular bifurcations, the daughter vessels of a venule continues or branches from the parent vessel at higher angles for a venule than for an arteriole, so that the mean angle between the daughter branches of a venular bifurcation is significantly smaller than the mean angle between the daughter branches of an arteriolar node. Note also that the mean number of nodes by arborescence is nearly 50% higher for veins than for arterioles and that, on average, arterioles divide more asymmetrically than veins.

The feed continuation angle increases with the order of the parent vessel, as well as the angular asymmetry, while the angle between daughter branches decreases.

The relationships between the area ratios and the parent vessel order are a little bit more complex: while the major AR and the asymmetry ratios increase (relative to parent vessel order), the minor AR decreases and these opposite trends result in a non-monotonic variation of the AER.

The relationships between the branching patterns and the bifurcation type are less straightforward even when the differences in parameter values for the three types are obvious consequences of the definition. This classification reflects differences in the OAI, i.e., differences between the diameters of the parent vessel and the smaller daughter branch. From a functional point of view, these last differences should generate differences in the phase separation effect (Pries et al., 1989) and thus in the hematocrit distribution at each bifurcation and, by a cumulative effect, in the entire network.

Most of the changes in the area ratios can be interpreted qualitatively by looking at the order differences between the parent and daughter vessels. For instance, the major daughter branch of an IEB or an ALB bifurcation, which has the same order as the parent vessel, essentially arises without tapering in the case of an IEB, or with a weak taper in the case of an ALB, whereas the major branch of an HSB, with a lower order than that of the parent vessel, undergoes a marked taper. The minor area ratio decreases as the OAI, that is to say the difference between the orders of the parent vessel and the minor daughter branch, increases. So, the AER, the sum of the major and minor branching ratios, results in the greatest average value for an IEB bifurcation (1.71) while the smaller taper of the major branch balances the higher size reduction of the minor branch of an ALB, with respect to an HSB type, to give approximately the same AER (1.50) for both types. Both daughters of an IEB or an HSB bifurcation have the same order while the order difference between the daughters for an ALB is at least 1: this latter type is therefore markedly more asymmetric (decreased asymmetry ratio) than the other two. This last observation cannot be extrapolated directly from the mere definition of this classification. Moreover, within each type group, the area ratios are modulated by the parent vessel order. With regard to the angles, the most significant difference observed as a function of the bifurcation type is the feed continuation angle of the ALB bifurcation type, which is higher than the corresponding angle of the other types. Within each type group, the modulations by the parent vessel order are very similar and correspond to the overall behavior with the taxonomy previously described.

The major empirical information yielded concerning the connectivity, derived from the study of the connectivity matrix, is that, on average, 4 parent vessels of a given order give rise to at least 3 daughter vessels of the same order and to 5 daughter vessels of a lower order.

3.2. *Limits of the study*

Possible limitations of the present study may result from three factors: the limited depth of the material used, the shrinkage effect and the postmortem change of vascular tone.

The depth of the material available for this study does not appear to confer significant limitations upon it since branching parameters, such as area ratios and angles, are computed from local data and are independent of the completeness of the vascular trees, at the difference of other parameters, such as cumulative length or volume, etc., analyzed in previous studies based on the same database.

Anatomical preparation is likely to induce shrinkage, generating an underestimation of morphometric data. However, uniform local shrinkage would seem to allow the result here to hold, regardless of more global effects of shrinkage. Injection of a low-viscosity fluid (a mixture of Indian-ink and gelatine) via the six cerebral arteries gives a complete and physiological-like perfusion of the whole vascular system of the brain. This technique, which was used to “mark” the vessels, is more likely to generate uniform shrinkage in the specimen compared to more sophisticated, newer methods (Heinzer et al., 2006; Mittal et al., 2005) that combine a number of different techniques such as casting – under high-pressure injection of highly viscous polymer – for large vessels or cell staining of histological sections for smaller ones.

Finally, a third factor, and one that influences any postmortem vascular study, concerns the fact that the diameters measured in the postmortem brain may not be the same as those found in intact vessels of a living brain due to the change in vascular tone. However, the pressure drops through a microvascular bifurcation are so small that this factor is unlikely to change significantly the ratios and angles measured locally.

3.3. *New insights for structural organization of the brain microvasculature and functional implications*

To our knowledge, this is the first study to generate and analyze so large a quantity of data, and the first to allow a sound, statistically grounded set of conclusions on the branching patterns of microvessels of the human cerebral cortex.

The choice of the diameter-defined Strahler taxonomy, first proposed by Fung and Kassab, responded to the objective of the morphometry of the tree, which was to use the data for

haemodynamics (Fung p 447) (Fung, 1996). Because, according to Poiseuille's formula, flow varies with the fourth power of the diameter and the first power of the length, they assigned priority to diameter. This taxonomy enabled these authors and many others, including those of the present paper, for human cerebral cortex, to demonstrate one of the main structural properties of vascular tree-like networks for most organs: their fractal nature. In a recent paper, we were able to derive the well-established allometric equation from this structural property and a structure–function relationship between flow and the vascular volume distal to a bifurcation (West et al., 1997), and hence to confirm, once again, the functional relevance of this taxonomy for the microcirculation as previously demonstrated by a significant body of work from that group.

3.3.1. *Branching ratios*

In recent years, Kassab and co-workers have devoted a series of papers to the morphometric data of porcine coronary microvascular networks (Kaimovitz et al., 2005; Kaimovitz et al., 2008; Kassab et al., 1993; Kassab, 2006; Mittal et al., 2005). In spite of the impressive database gathered by Kassab and co-workers on these networks, that study contains little detailed information on branching patterns that would have allowed comparison with the findings in the present study. Moreover, the comparisons are problematic as the branching ratios in their papers are generally defined as the ratios of diameters while we have considered area ratios: given that the distributions are skewed, the mean values in the one study are not comparable with the squares in the others. Moreover, the asymmetry ratio is defined as a ratio of small to large daughters in the present work and inversely by Kassab's group; the comparisons are all the more difficult since no clear statistical characterization of the distribution functions are given in Kassab et al. These difficulties may explain some discrepancies, for instance why the AER, which increases incrementally towards order 1 for porcine coronary vessels, might vary non-monotonically for human cerebral microvessels?

Regarding this last parameter, further discussion is needed. Indeed, according to the theoretical prediction derived from Murray's Law (Zamir et al., 1979), the maximal attainable value of AER should be 1.26, which has in the past been considered as an "optimal" value, at least for symmetrical bifurcations (Zamir, 1977). However, for nearly three-quarters of the cerebral bifurcations considered in this study, the measured AER is higher than this "optimal" value. In addition, for all groups considered, whatever the vessels nature, topology or type, the mean AER are either higher or close to 1.5. By contrast, in the porcine coronary arterial tree, the AER has a value close to unity in vessels with an order greater than 5 and a global mean value, i.e., when the entire arterial tree is considered, of 1.12 (VanBavel and Spaan, 1992). However, for the lower orders of porcine coronary vessels (Kaimovitz et al., 2008), the AER are slightly lower than 1.5, increasing towards this value with decreasing orders. Thus, our findings for human cerebral microvessels are in qualitative agreement with the findings of Kaimovitz for the lower orders of pig coronary vessels (Kaimovitz et al., 2008).

To determine the hemodynamic consequences of bifurcation asymmetry and its functional implications, Kaimovitz et al. (2008) derived from structure–structure and structure–function relationships (scaling laws) the approximate relations: $\frac{R_1^c}{R_2^c} = \frac{A_2}{A_1}$ (K8) and $\frac{Q_1}{Q_2} = \left(\frac{A_2}{A_1}\right)^{-1}$ (K9) (transposed from Eqs. (A8) and (A9) in Appendix of Kaimovitz's paper using the notations of the present paper), where R_1^c and R_2^c are the equivalent resistances of distal tree to large and small daughter vessels, respectively. Eq. (K8) implies that a large daughter vessel has a smaller crown resistance, and vice versa. Eq. (K9) models the ratio of flow (Q_1/Q_2) as being smaller at lower orders and increasing at higher orders so that the distribution in flow at bifurcations of lower order (perfusion) vessels becomes more uniform, suggesting that the distal microvasculature may play a regulatory role in the arterial tree.

The AER increases toward vessel order 1 (Fig. 4, lower left) and both the asymmetry ratio and flow ratio tend towards unity as vessel order tends to 1. If we assume then that $U_1 \approx U_2$, where U_1 and U_2 are the velocity in the large and small daughter branches, it is easy to show that $\frac{U_0}{U_1}$ = AER (see Eq. (A12) in Appendix of Kaimovitz's paper), where U_0 is the velocity in the parent vessel, and thus that the velocity rapidly decreases from higher to lower orders.

For a Murray-type relation, the flow in any branch behaves as a power-law of the diameter of this branch, so $Q \sim D^k$. The wall shear stress, τ, can be related to the exponent k: $\tau = \frac{32\mu Q}{\pi D^3}$, namely $\tau \sim D^{k-3}$, assuming, as a first and crude approximation, a constant blood viscosity μ. If k tends to 3 (Murray's law) as it does, approximately, for larger vessels (those of a higher order), the wall shear stress is uniform over different sized vessels. If k tends to 4, as it does (approximately) for smaller vessels, the shear stress decreases with a decrease in diameter, i.e., reduces near the capillary net. The variation in k for different size vessels is shown in Fig. 10 and may reflect the transitions for changes in shear stress. However, as shown in Fig. 10, the median value of the exponent k varies also with the type of bifurcation, more precisely increases, from approximately 3 to above 4, with the OAI, namely with the difference of diameters between the parent vessel and the smaller daughter branch. Moreover, taking into consideration the phase separation effect that induces significant hematocrit and blood viscosity variations and the huge statistical dispersion of the exponent k, interpretation does require a degree of caution. A more reliable understanding of the functional implications of our findings on the branching patterns would require further full analysis based on a complete mathematical treatment on a realistic microvascular network including these morphometric data.

3.3.2. *Branching angles*

Zamir et al. (1983) pointed out that measurements of branching angles, whether in human or animal tissue, are not available in adequate quantity. Furthermore, nearly all previous measurements were derived from two-dimensional pictures of the vascular bed and were hence subject to three-dimensional viewing errors (Zamir et al., 1983). These operators were the first to actually measure the degree of three-dimensionality of each bifurcation using dedicated viewing equipment. However, these measurements (of branching angles) were taken manually – an average of measurements made by two operators – which limited both the total number of measurements (500 on arterial bifurcations of rats) as well as their accuracy. Taking into account

"the difficulties involved in measuring angles accurately, and allowing for a reasonable amount of biological scatter", they concluded that arterial bifurcations are mostly two-dimensional. This conclusion was supported in a recent model of coronary networks which described a planar configuration of bifurcations (Kaimovitz et al., 2005). However, this conclusion should be moderated in view of the results obtained both by Zamir and by the present study: 12% (respectively 28%) of the arteriolar bifurcations measured in rat have an out-of-plane angle greater than 10° (respectively 5°), and the histogram of this angle (see Fig. 5 in Zamir) (Zamir et al., 1983) displays a striking similarity with the hyperbolic-shaped distribution highlighted in the present study. This finding should come as no surprise: that the bifurcations of human cerebral cortex, an organ with a richly three-dimensional vascular structure, should show a more marked degree of three-dimensionality than the arteriolar bifurcations seen in the cardiovascular system of the rat, containing largely two-dimensional vascular networks. Our findings also concur with important out-of-plane features found in some human cerebral bifurcations (distal middle cerebral artery) (Zakaria et al., 2008). Zamir et al. (1983) notes that in an arterial bifurcation the larger branch makes a smaller angle with the direction of the parent artery than does the smaller branch, a finding that is in clear agreement with our findings here, once one takes into account that, in Zamir's work, the branching angles between the parent vessel and the daughter branches refers to the prolongation of the mother branch, and not the parent vessel itself. Moreover, Zamir notes that this rule is not followed rigidly, as demonstrated clearly in the polar histograms (Fig. 5) in the present study.

Various studies by Frame and Sarelius (1993, 1995) have shown that arteriolar bifurcation angle varies with position downstream of the feeding arteriole. With respect to this initial work on a limited number ($N=181$) of bifurcations in hamster cremaster muscle arterioles, our findings clearly confirm this finding for the angles between the mother vessel and the larger daughter vessel and between both daughter branches, but not for the angles between the parent vessel and the smaller daughter, which do not vary significantly with position. Frame and Sarelius also calculated theoretical angles of bifurcation from the asymmetry ratio (γ), assuming total energy minimization, and proposed the following relationship: $\cos\theta = \frac{\left(1+\gamma^{3/2}\right)^{4/3}-\left(1+\gamma^{2}\right)}{2\gamma}$, where θ refers to the angle between both daughter branches. They compared predicted to measured angles. We applied the same calculations and comparisons to our data, arriving at the same conclusions as these authors: models based on optimality principles fail to predict the measured angles of bifurcation. More precisely, the correlation between predicted and measured angles is close to zero, measured angles are normally distributed around a value superior to 90° and spread over a wide range, while predicted angles concentrate on a very narrow range, between 75° and 83°, with a left-skewed distribution, and do not show positional variation.

3.3.3. *Optimal design of branching patterns*

Since the work of Murray in 1926, the discussion on the optimal design of vascular branching has been the motivation behind much work in this field (for a review, see Bennett et al., 2000). Optimality underpins the origins and assumptions wielded in many recent papers, including some in the important field of remodeling. However, a growing body of evidence suggests that this assumption of optimality within the circulatory system may not in fact be a valid one to make (Djonov et al., 2002). We have previously shown (Cassot et al., 2009), in work whose conclusions are supported by the detailed analysis presented here, that the bifurcation exponent, derived from a generalized form of Murray's law, shows so large a spread so as to lead us to the conclusion that such a law (and perhaps the optimal design principle that underpins it) is ultimately unable to account for the true complexity found within the architecture of microvascular branching. Without questioning universal bifurcation theories, Bennett et al. (2000) have demonstrated the perinatal pulmonary arterial circulation to be a clear exception to these theories of vessel form and function. Moreover, they recapitulated the premises upon which these universal theories are based and showed particularly how principles of economy underlie them. Many of these studies were published at the beginning of the third quarter of the last century. Interestingly, in 1997, Benoit Mandelbrot (1998), the father of fractals wondered whether "au troisième quart de ce siècle, la recherche de l'unifié, de l'abstrait et du general dominait la recherche" (in the third quarter of this century, the quest towards unification, abstraction and generality dominated research). Mandelbrot proposed a new general structure to underlie many apparently heterogeneous phenomena: the concept of scale invariance or self-similarity. We and others have shown that branching vascular networks are ruled by such scaling laws and are, as a result, fractal. However, it is far beyond the scope of this paper to conclude whether the concepts of fractal geometry may lead to a more adequate theoretical interpretation of microvascular networks branching patterns than optimality principles have thus far managed.

At last, optimality deviance seems to have some clinical implications. Tapp et al. (2007) studied retinal microvascular abnormalities in a population of children to determine whether retinal microvascular abnormalities are associated with early life factors. They showed that small birth size may be associated with structural variations in the retinal vasculature in children especially with optimality deviance of retinal bifurcations and retinal tortuosity. They concluded that these are important observations that could further our understanding of mechanisms linking the long-observed association among low birth weight and cardiovascular disease (increased risk of stroke, ischemic heart disease), hypertension, and diabetes in adults.

Witt et al. (2006) examined the relationship between abnormalities of the retinal microvasculature and death from ischemic heart disease (IHD) and stroke and showed that increased risk of IHD death was associated with a suboptimal relationship of arteriolar diameters at bifurcation, i.e., a deviation from optimal design.

3.4. *Conclusion*

Apart from a theoretical perspective on bifurcation design, the main thrust of the present study is to provide a complete data set of branching patterns, with their statistical distributions

and relationships to vessel nature, order or type. It completes the morphometric and topological data on segments length and diameter which have previously been derived from our database together with scaling laws (Cassot et al., 2009). Branching patterns and 3D geometry of the cortical vasculature are needed to carry out a large-scale reconstruction of the cortical vascular trees to allow for future studies of hemodynamics. Algorithms for such reconstructions, based on detailed anatomical data, have been applied to the porcine coronary arterial vasculature (Kaimovitz et al., 2005) and to the cerebral macrocirculation (Schreiner et al., 2006).

Indeed realistic three-dimensional representations of the microvascular structure are necessary to construct models which integrate the available data into a functional description of the physiology (Beard, 2001), especially when, for example, microvascular networks do not lend themselves well to the Krogh description, which is certainly the case for the cerebral cortex microcirculation (Secomb et al., 2000). It may be of some relevance to recall that, as first suggested by Bassingthwaighte (2000), the primary goal in this endeavor is the construction of the databases and that this step necessarily comes before the secondary goal, namely the construction of models based on such databases, even when the exercise of working towards this last goal may feed back information that refines the primary goal (Beard, 2001). This logical requirement must be specially fulfilled when the anatomical database corresponding to the organ considered is at an early stage of development, as it is for the brain (Hunter and Nielsen, 2005).

In this paper, we considered only some potential implications for the structure and function of the cerebral microcirculation of the morphometric data on branching patterns presented here. However, though they do bring new insights to cerebral microcirculation function, they should be considered only as general trends. Further efficient simulations, based on realistic morphometric data (i.e., working towards the Bassinghwaighte secondary goal; Bassingthwaighte, 2000), are still necessary to derive more precise and well-documented information on these functional properties.

4. Experimental procedures

4.1. *Creation of a microvascular network database of the human cerebral cortex*

The procedures used for image acquisition, mosaic construction and vessels segmentation have been described in detail elsewhere (Cassot et al., 2006) and are outlined briefly below.

4.1.1. *Data acquisition*

Digital three-dimensional images of the network were obtained from thick sections (300 μm) of India ink-injected human brain (Duvernoy et al., 1981) by confocal laser microscopy. The specimen used in this study was a 60-year-old female who died from abdominal lymphoma without any known vascular or cerebral disease. The secondary cortex next to the collateral sulcus in the right temporal lobe was the anatomical region chosen for this study owing to the outstanding quality of the injection in this area. The entire volume of the cortex on both sides of the sulcus (fusiform and parahippocampal gyri) was digitized on 3 adjacent coronal sections. Each elementary block contained 70 sections of 512×512 pixels, with a voxel size of 1.22×1.22×3 μm.

These individual bricks covering a large zone were then added in a special data object called an image mosaic and were realigned ensuring that the vessels in the overlapping area matched perfectly. Ten mosaics were so constructed to cover the entire cortex area chosen corresponding to a cortical volume of 28.6 mm^3.

4.1.2. *3D microvascular network segmentation and extraction of tree-like vessels*

Each image mosaic contains a large quantity of data, typically several gigabytes and cannot be loaded and processed at once in the memory of a standard personal computer. Adapted methods were developed to adequately process these images (Cassot et al., 2006; Fouard et al., 2006). The main feature and advantage of these methods is that they process data locally while preserving global properties. This enabled extraction of vessel center lines and computation of distance maps which measure the distance between any point inside a vessel to the nearest vessel wall, providing for each center line point a good estimation of the vessel radius. Finally, a skeleton of the 3D microvascular network was obtained for each mosaic and a value of the vessel radius estimated from the distance map was attached to every vertex. For further processing of the morphometric data and analysis of the topology, the lineset was stored in an ASCII file, each line in this file corresponding to a center line point and its *x*, *y*, *z* coordinates, as well as the radius of the vessel at this point.

Next, to account for the dual nature of this network, we separated the tree-like arterioles and venules from the capillary network. To extract a vascular tree, we identified its origin in the sulcus and traced all the paths from this origin through vessels in which resistance to flow is lower than a prescribed value (Cassot et al., 2006).

4.2. *Vessel taxonomy and type*

These vascular trees were analyzed according to the diameter-defined Strahler taxonomy (Cassot et al., 2006; Kassab, 2006) (see Fig. 11). This taxonomy produced a hierarchical ranking of vessel elements assigning them an order from 0, for the "terminal" branches connected to the capillaries, to a maximal rank for the branches closest to the pial tree origin. The manner in which vessels of a given order are connected to vessels of another order can be represented by a connectivity matrix (see below). In addition, for classification purposes, we defined the Branching Development Index (BDI), equal to the total number of orders in the arborescence, and three bifurcation types:

- the intra-element bifurcations (IEB) in which the three branches have the same order;
- the homogeneous-symmetrical bifurcations (HSB) in which the order of the two daughter branches equals that of the parent vessel minus one; and

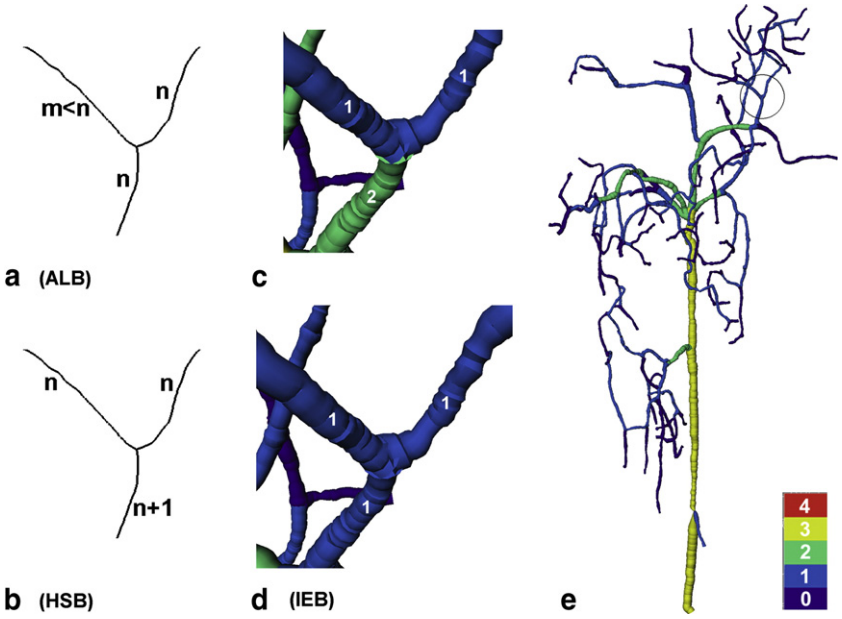


Fig. 11 – Taxonomy and bifurcation types: illustration of the Strahler classification scheme and definition of ALB (a) and HSB (b) types. The diameter-defined Strahler taxonomy (d) modifies the Strahler classification (c) when the diameter of the confluent segment is smaller than $[(Dn+SDn)+(Dn+1-SDn+1)]/2$, where Dn and SDn denote the mean and standard deviation of the diameters of the vessels of order n, leading then to an IEB bifurcation. Example of the topology obtained with this taxonomy for an arteriolar arborescence (e).

- the asymmetric-lateral bifurcations (ALB) in which a lower order vessel branches laterally between two segments of the same order.

In the following sections, these types will be classified in descending order of their Order Asymmetry Index (OAI), i.e., the difference between the parent vessel order and the smaller daughter branch order (type 1: ALB, OAI > 1; type 2: HSB, OAI = 1; type 3: IEB, OAI = 0).

4.3. *Connectivity matrix*

The normalized connectivity matrix is defined as follows (Turcotte et al., 1998): its element (i,j) is called the connectivity ratio and is calculated as the ratio of the total number of daughter vessels of order i, issued from a parent vessel of order j, to the total number of parent vessels of order j. The connectivity matrix is an upper triangular matrix since the connectivity ratio is null when $j<i$. Therefore, the sum of each column equals 2 since the total number of daughter vessels of order $i \leq j$ issued of a parent vessel of order j is twice the number of parent vessels of order j.

4.4. *Derivation of geometrical parameters*

In order to remove the possibility of spurious local curvature discontinuity of the vessel, which may result from an artifact of voxel size or from the segmentation process, center line coordinate data of each vascular segment were smoothed prior to 3D geometrical analysis. This was achieved using a simple five-point weighted local smoothing, centered on each vertex, applied to each coordinate of the center line points. Arc length s was then determined at each vertex. This smoothed curve description was then resampled using cubic spline interpolation to give points at constant 0.5-μm intervals along the curve length. First derivatives, with respect to the arc length, were simultaneously derived for each coordinate of the resampled center line points of the smoothed curve. This smoothing and interpolating process yielded well-defined parameterized center line curves for each vascular segment in the form $r(s)=(x(s), y(s), z(s))$ with their corresponding derivatives $r'(s)=(x'(s), y'(s), z'(s))$.

4.5. *Branching angles*

A node or bifurcation point is a point common to three center line curves. It is located inside three vascular segments. Instead of computing the local branching angles at the node, we used the boundaries of the intersecting vessels as a more elegant point of computation, as illustrated in Fig. 12. More precisely, for each segment i connected to the node, we identified the last vertex inside and the first vertex outside a sphere centered at the node which radius equals the maximal local radius of the three vessels (see Fig. 12) and took the mean of the tangential unit vectors at both points. Let u_i be the unit vector obtained this way, where $i=0$, 1 or 2 points out, respectively, the mother, major and minor daughter branch of the node. The branching angles were defined as $\alpha_{ij}=\cos^{-1}(u_i \cdot u_j)$. The asymmetry angle was defined as $\beta=\alpha_{01}-\alpha_{02}$.

The three branches of a bifurcation define a solid angle Ω equal to the sum of the angles between the three branches: $\Omega=\alpha_{01}+\alpha_{02}+\alpha_{12}-\pi$. This angle is less than π and equals this value only when the three branches are co-planar. The value

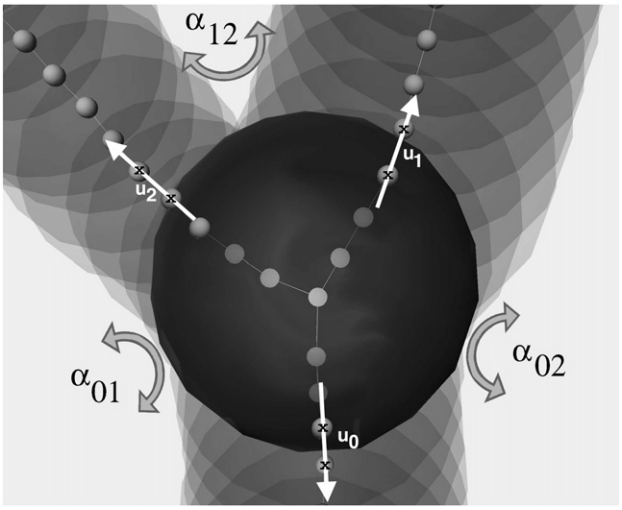


Fig. 12 – Three-dimensional representation of a bifurcation illustrating the method used to compute the bifurcation angles (see text).

of $\pi-\Omega$, which is called the out-of-plane angle, was used to define the planarity of the bifurcation.

4.6. *Branching area ratios*

Important geometric features of a bifurcation are the branching area ratios. We analyzed the total area ratio ($(A_1+A_2)/A_0$), sometimes called the area expansion ratio (AER), the major (A_1/A_0) and the minor (A_2/A_0) area ratios as well as the asymmetry ratio ($\gamma=A_2/A_1$), where A_0, A_1 and A_2 are the cross-sectional areas of the corresponding branches and where the daughter branches are chosen such that $D_1>D_2$. Note that some authors (Kaimovitz et al., 2008) focus on the ratio of diameters. However, the area ratios have a direct significance regarding the dynamics of blood flow because the vessel areas (and not their diameters) are the relevant quantities for mass conservation.

4.7. *Optimality of branching patterns (Is bifurcation design optimal?)*

In order to analyze the validity of the generalized Murray's law ($d_0^k=d_1^k+d_2^k$), we derived the branching exponent for every bifurcation, using an iterative algorithm, and analyzed its statistical distribution for the 9414 bifurcations and for each group defined on the basis of the vascular nature, taxonomy or type.

4.8. *Statistical analysis*

A standard statistical analysis was applied to scalar variables (area ratios, branching exponents). Specific statistical methods were applied to the distributions of directional data (branching angles): each angle was represented as a unit vector (r_i) and the mean angle of a population was defined from the direction of the resulting vector $R = \sum_1^N r_i$, while the circular standard deviation (CSD) was defined according to $CSD = \frac{180^\circ}{\pi}\sqrt{-2\ln R}$ (Zar, 1984).

To determine statistical significance of differences, especially between the mean or median values of the branching parameters as a function of the nature, taxonomy or bifurcation type of the bifurcations, we carried out various statistical analyses, including unpaired Kruskal–Wallis and Wilcoxon–Mann–Whitney non-parametric tests for branching ratios. Watson's U^2 two-sample tests with ties – taking into account that there are two or more observations with the same numerical value – were applied to test the null hypothesis for two mean angles (Zar, 1984). The Watson U^2 test compared the critical values of U^2_{P,n_1n2}, where n_1 and n_2 are the two-sample sizes and P the significance level (Zar, 1984, Table B38; Appendix B—App195-7 was used to yield P). In the majority of the tests, both sample sizes were greater than 100 and the critical values for sample sizes of infinity were computed by $U^2_{P,\infty,\infty} = -\left(\frac{1}{2\pi^2}\right)\left[\ln\left(\frac{P}{2}\right) - \ln\left\{1+\left(\frac{P}{2}\right)^3\right\}\right]$, from which the P-value could be readily inferred.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.brainres.2009.12.007.

REFERENCES

Baish, J.W., Jain, R.K., 2000. Fractals and cancer. Cancer Res. 60, 3683–3688.

Baish, J.W., Gazit, Y., Berk, D.A., Nozue, M., Baxter, L.T., Jain, R.K., 1996. Role of tumor vascular architecture in nutrient and drug delivery: an invasion percolation-based network model. Microvasc. Res. 51, 327–346.

Bassingthwaighte, J.B., 2000. Strategies for the physiome project. Ann. Biomed. Eng. 28, 1043–1058.

Beard, D.A., 2001. Computational framework for generating transport models from databases of microvascular anatomy. Ann. Biomed. Eng. 29, 837–843.

Bennett, S.H., Eldridge, M.W., Zaghi, D., Zaghi, S.E., Milstein, J.M., Goetzman, B.W., 2000. Form and function of fetal and neonatal pulmonary arterial bifurcations. Am. J. Physiol., Heart Circ. Physiol. 279, H3047–H3057.

Bullitt, E., Muller, K.E., Jung, I., Lin, W., Aylward, S., 2005a. Analyzing attributes of vessel populations. Med. Image Anal. 9, 39–49.

Bullitt, E., Zeng, D., Gerig, G., Aylward, S., Joshi, S., Smith, J.K., Lin, W., Ewend, M.G., 2005b. Vessel tortuosity and brain tumor malignancy: a blinded study. Acad. Radiol. 12, 1232–1240.

Bullitt, E., Lin, N.U., Smith, J.K., Zeng, D., Winer, E.P., Carey, L.A., Lin, W., Ewend, M.G., 2007. Blood vessel morphologic changes depicted with MR angiography during treatment of brain metastases: a feasibility study. Radiology 245, 824–830.

Carmeliet, P., 2005. Angiogenesis in life, disease and medicine. Nature 438, 932–936.

Carmeliet, P., Jain, R.K., 2000. Angiogenesis in cancer and other diseases. Nature 407, 249–257.

Cassot, F., Lauwers, F., Fouard, C., Prohaska, S., Lauwers-Cances, V., 2006. A novel three-dimensional computer-assisted method for a quantitative study of microvascular networks of the human cerebral cortex. Microcirculation 13, 1–18.

Cassot, F., Lauwers, F., Lorthois, S., Puwanarajah, P., Duvernoy, H., 2009. Scaling laws for branching vessels of human cerebral cortex. Microcirculation 16, 331–346.

Djonov, V.G., Kurz, H., Burri, P.H., 2002. Optimality in the developing vascular system: branching remodeling by means of intussusception as an efficient adaptation mechanism. Dev. Dyn. 224, 391–402.
Duvernoy, H.M., Delon, S., Vannson, J.L., 1981. Cortical blood vessels of the human brain. Brain Res. Bull. 7, 519–579.
Fouard, C., Malandain, G., Prohaska, S., Westerhoff, M., 2006. Blockwise processing applied to brain microvascular network study. IEEE Trans. Med. Imaging 25, 1319–1328.
Frame, M.D., Sarelius, I.H., 1993. Arteriolar bifurcation angles vary with position and when flow is changed. Microvasc. Res. 46, 190–205.
Frame, M.D., Sarelius, I.H., 1995. Energy optimization and bifurcation angles in the microcirculation. Microvasc. Res. 50, 301–310.
Fukumura, D., Jain, R.K., 2008. Imaging angiogenesis and the microenvironment. Apmis 116, 695–715.
Fung, Y.C., 1996. Biomechanics : Circulation, Springer, New York.
Goense, J.B., Zappe, A.C., Logothetis, N.K., 2007. High-resolution fMRI of macaque V1. Magn. Reson. Imaging 25, 740–747.
Heinzer, S., Krucker, T., Stampanoni, M., Abela, R., Meyer, E.P., Schuler, A., Schneider, P., Muller, R., 2006. Hierarchical microimaging for multiscale analysis of large vascular networks. NeuroImage 32, 626–636.
Heinzer, S., Kuhn, G., Krucker, T., Meyer, E., Ulmann-Schuler, A., Stampanoni, M., Gassmann, M., Marti, H.H., Muller, R., Vogel, J., 2008. Novel three-dimensional analysis tool for vascular trees indicates complete micro-networks, not single capillaries, as the angiogenic endpoint in mice overexpressing human VEGF(165) in the brain. NeuroImage 39, 1549–1558.
Hunter, P., Nielsen, P., 2005. A strategy for integrative computational physiology. Physiology (Bethesda) 20, 316–325.
Jain, R.K., 2005. Normalization of tumor vasculature: an emerging concept in antiangiogenic therapy. Science 307, 58–62.
Jonk, A.M., Houben, A.J., de Jongh, R.T., Serne, E.H., Schaper, N.C., Stehouwer, C.D., 2007. Microvascular dysfunction in obesity: a potential mechanism in the pathogenesis of obesity-associated insulin resistance and hypertension. Physiology (Bethesda) 22, 252–260.
Kaimovitz, B., Lanir, Y., Kassab, G.S., 2005. Large-scale 3-D geometric reconstruction of the porcine coronary arterial vasculature based on detailed anatomical data. Ann. Biomed. Eng. 33, 1517–1535.
Kaimovitz, B., Huo, Y., Lanir, Y., Kassab, G.S., 2008. Diameter asymmetry of porcine coronary arterial trees: structural and functional implications. Am. J. Physiol., Heart Circ. Physiol. 294, H714–H723.
Kassab, G.S., 2006. Scaling laws of vascular trees: of form and function. Am. J. Physiol., Heart Circ. Physiol. 290, H894–903.
Kassab, G.S., Rider, C.A., Tang, N.J., Fung, Y.C., 1993. Morphometry of pig coronary arterial trees. Am. J. Physiol. 265, H350–H365.
Lauwers, F., Cassot, F., Lauwers-Cances, V., Puwanarajah, P., Duvernoy, H., 2008. Morphometry of the human cerebral cortex microcirculation: general characteristics and space-related profiles. NeuroImage 39, 936–948.
Levy, B.I., Schiffrin, E.L., Mourad, J.J., Agostini, D., Vicaut, E., Safar, M.E., Struijker-Boudier, H.A., 2008. Impaired tissue perfusion: a pathology common to hypertension, obesity, and diabetes mellitus. Circulation 118, 968–976.
Lu, M., Zhang, Z.G., Chopp, M., 2004. Analysis of cerebral microvascular architecture—application to cortical and subcortical vessels in rat brain. J. Neurosci. Methods 138, 81–87.
Mandelbrot, B., 1998. Fractales, hasard et finance, Flammarion, Paris.
Meyer, E.P., Ulmann-Schuler, A., Staufenbiel, M., Krucker, T., 2008. Altered morphology and 3D architecture of brain vasculature in a mouse model for Alzheimer's disease. Proc. Natl. Acad. Sci. U. S. A. 105, 3587–3592.
Mittal, N., Zhou, Y., Linares, C., Ung, S., Kaimovitz, B., Molloi, S., Kassab, G.S., 2005. Analysis of blood flow in the entire coronary arterial tree. Am. J. Physiol.. Heart Circ. Physiol. 289, H439–H446.
Murray, C.D., 1926. The physiological principle of minimum work. I. The vascular system and the cost of blood volume. Proc. Natl. Acad. Sci. U. S. A. 12, 207–214.
Murfee, W.L., Schmid-Schonbein, G.W., 2008. Chapter 12. Structure of microvascular networks in genetic hypertension. Methods Enzymol. 444, 271–284.
Nishimura, N., Schaffer, C.B., Friedman, B., Lyden, P.D., Kleinfeld, D., 2007. Penetrating arterioles are a bottleneck in the perfusion of neocortex. Proc. Natl. Acad. Sci. U. S. A. 104, 365–370.
O'Flynn, P.M., O'Sullivan, G., Pandit, A.S., 2007. Methods for three-dimensional geometric characterization of the arterial vasculature. Ann. Biomed. Eng. 35, 1368–1381.
Patton, N., Aslam, T., Macgillivray, T., Pattie, A., Deary, I.J., Dhillon, B., 2005. Retinal vascular image analysis as a potential screening tool for cerebrovascular disease: a rationale based on homology between cerebral and retinal microvasculatures. J. Anat. 206, 319–348.
Pries, A.R., Ley, K., Claassen, M., Gaehtgens, P., 1989. Red cell distribution at microvascular bifurcations. Microvasc. Res. 38, 81–101.
Pries, A.R., Reglin, B., Secomb, T.W., 2005. Remodeling of blood vessels: responses of diameter and wall thickness to hemodynamic and metabolic stimuli. Hypertension 46, 725–731.
Risser, L., Plouraboue, F., Steyer, A., Cloetens, P., Le Duc, G., Fonta, C., 2007. From homogeneous to fractal normal and tumorous microvascular networks in the brain. J. Cereb. Blood Flow Metab. 27, 293–303.
Schreiner, W., Karch, R., Neumann, M., Neumann, F., Szawlowski, P., Roedler, S., 2006. Optimized arterial trees supplying hollow organs. Med. Eng. Phys. 28, 416–429.
Secomb, T.W., Hsu, R., Beamer, N.B., Coull, B.M., 2000. Theoretical simulation of oxygen transport to brain by networks of microvessels: effects of oxygen supply and demand on tissue hypoxia. Microcirculation 7, 237–247.
Tapp, R.J., Williams, C., Witt, N., Chaturvedi, N., Evans, R., Thom, S.A., Hughes, A.D., Ness, A., 2007. Impact of size at birth on the microvasculature: the Avon Longitudinal Study of Parents and Children. Pediatrics 120, e1225–e1228.
Turcotte, D.L., Pelletier, J.D., Newman, W.I., 1998. Networks with side branching in biology. J. Theor. Biol. 193, 577–592.
VanBavel, E., Spaan, J.A., 1992. Branching patterns in the porcine coronary arterial tree. Estimation of flow heterogeneity. Circ. Res. 71, 1200–1212.
Vanzetta, I., 2006. Hemodynamic responses in cortex investigated with optical imaging methods. Implications for functional brain mapping. J. Physiol. Paris 100, 201–211.
Weber, B., Keller, A.L., Reichold, J., Logothetis, N.K., 2008. The microvascular system of the striate and extrastriate visual cortex of the macaque. Cereb. Cortex 18, 2318–2330.
West, G.B., Brown, J.H., 2005. The origin of allometric scaling laws in biology from genomes to ecosystems: towards a quantitative unifying theory of biological structure and organization. J. Exp. Biol. 208, 1575–1592.
West, G.B., Brown, J.H., Enquist, B.J., 1997. A general model for the origin of allometric scaling laws in biology. Science 276, 122–126.
Wiernsperger, N., Nivoit, P., De Aguiar, L.G., Bouskela, E., 2007. Microcirculation and the metabolic syndrome. Microcirculation 14, 403–438.
Witt, N., Wong, T.Y., Hughes, A.D., Chaturvedi, N., Klein, B.E., Evans, R., McNamara, M., Thom, S.A., Klein, R., 2006. Abnormalities of retinal microvascular structure and risk of mortality

from ischemic heart disease and stroke. Hypertension 47, 975–981.
Zakaria, H., Robertson, A.M., Kerber, C.W., 2008. A parametric model for studies of flow in arterial bifurcations. Ann. Biomed. Eng. 36, 1515–1530.
Zamir, M., 1977. Shear forces and blood vessel radii in the cardiovascular system. J. Gen. Physiol. 69, 449–461.
Zamir, M., Medeiros, J.A., Cunningham, T.K., 1979. Arterial bifurcations in the human retina. J. Gen. Physiol. 74, 537–548.
Zamir, M., Wrigley, S.M., Langille, B.L., 1983. Arterial bifurcations in the cardiovascular system of a rat. J. Gen. Physiol. 81, 325–335.
Zar, H., 1984. Biostatistical Analysis, Prentice-Hall, Englewood Cliffs, NJ.